```
//Program on switch as an expression :
-----------------------------------------
 //switch as an expression
 //WAP to verify whether the given day is weekday OR weekend
 void main()
 {
      int choice = Integer.parseInt(IO.readln("Enter your choice 1 - 7 :"));

      String result = switch(choice)
      {
       case 1,2,3,4,5 -> "Weekday";
         case 6,7 -> "Weekend";
         default -> "Invalid Day";
      };
      IO.println("Today is, Day "+choice+" so, It is a "+result);
 }
```

```
 SwitchDemo6.java
 --------------------
 /* Employee performer of the year
   Grade A -> High-level performer [bonus amount is 20% of the salary]
   Grade B -> Mid-level performer [bonus amount is 15% of the salary]
   Grade C -> Basic level performer [bonus amount is 10% of the salary]
   No grade -> 5 % of the salary
 */
 void main()
 {
      char grade = IO.readln("Enter employee grade :").toUpperCase().charAt(0);
      double salary = Double.parseDouble(IO.readln("Enter employee salary :"));

      double bonus = switch(grade)
      {
       case 'A' ->
           {
                IO.println("High-level performer");
                yield salary * 0.20;
           }
           case 'B' ->
           {
                IO.println("Mid-level performer");
                yield salary * 0.15;
           }
           case 'C' ->
           {
                IO.println("Basic-level performer");
                yield salary * 0.10;
           }

         default -> salary * 0.05;
      };

      IO.println("Employee Salary amount is : "+salary);
      IO.println("Employee bouns amount is : "+bonus);
      IO.println("Employee total salary is : "+(salary + bonus));
 }
```

```
 SwitchDemo7.java
 ---------------------
 void main()
 {
    long x = 12;  //long, float, double and boolean are not allowed

      switch(x)
      {
            case 12L:
      }
 }
```

```
  SwitchDemo8.java
  --------------------
  void main()
  {
       int x = Integer.parseInt(IO.readln("Enter the value of x :"));
       final int y = 10;

       switch(x)
       {
            case y -> IO.println("It is case 10");

            case 5 + 6 -> IO.println("It is case 11");
       }
  }
```

  SwitchDemo9.java

  While working with byte, short and char the **label** of the case must be within the **range** only otherwise we will get compilation error

```
  void main()
  {
       short b = 120;

       switch(b)
       {
            case 120 -> IO.println("Case 120");
            case 127 -> IO.println("Case 127");
            case 128 -> IO.println("Case 127");
            case 33000 -> IO.println("Case 127"); //error out of the range
       }
  }
```

2.Iteration OR Looping Statements :

Loops in Java:

If we want to execute **the same code** multiple times then writing the same code again and again is not a good choice.
Example:
I want to print “Welcome to Java” 10 times then writing this print statement 10 times is not a good programming practice.

```
Repeat.java
--------------
//Not a good choice
void main()
{
     IO.println("Welcome to Java");
     IO.println("Welcome to Java");
     IO.println("Welcome to Java");
     IO.println("Welcome to Java");
     IO.println("Welcome to Java");
     IO.println("Welcome to Java");
     IO.println("Welcome to Java");
     IO.println("Welcome to Java");
     IO.println("Welcome to Java");
     IO.println("Welcome to Java");
}
```

Here we should use looping statement:

Loops in Java are used to execute a **block of code multiple times** as long as the given Condition is **true.**
It also comes under control flow statement.
It helps us to reduce code duplication as well as to write the logic **once** and control how many times it runs using a condition.

Types of Loops in Java:
Java provides four types of Loops:

1) do while loop
2) while loop
3) for loop
4) for-each loop

Common Characteristics of loop:

a) Initialization (Starting point of the loop)
b) Condition (Ending point of the loop)
c) Updating (Increment/Decrement)

do-while loop

It is an **exit-controlled loop** because without checking the condition, it will execute the body at-least once.
Syntax:

```
do
 {
   // statements
 }
while(condition);
```

Programs :

```
 DoWhile1.java
 -------------
 void main()
 {
      do
      {
           int x = 1;  //x is local variable but accessible within the do
           IO.println(x);            //block only
           x++;
      }
      while (x<=10); //error
 }
```

```
 DoWhile2.java
 -------------
 void main()
 {
      int x = 1;
      do
      {
           IO.println(x);
           x++;
      }
      while (x<=10);
 }
```

```
 DoWhile3.java
 -------------
  void main()
  {
       int x = 1000;
       do
       {
            IO.println(x);  //1000   exit control behaviour
            x++;
       }
       while (x<=10);
  }
```

```
 DoWhileDemo4.java
 -----------------------
  void main()
  {
       int x = 0; //0
       do
     {
            IO.println(x);  //0 0
            x += x++;   //x  = x + x++;  //x = 0  + 0
     }
     while (x<=5);

  }
```

  Infinite loop, Output is 0

```
DoWhileDemo5.java
-----------------------
 void main()
 {
      int x = 1; //6
      do
    {
           IO.println(x); //1 2  4
           x += x++;   // x = x + x++;    x = 3  + 3
    }
    while (x<=5);

 }
```

```
 void main()
 {
      int i = 1;

     do
       {
      IO.println(i);
     }
      while (i++ <= 3);
 }
```

```
 -----------------------------------------------------------------------

 void main()
 {
      int i = 1;

    do;
      while (i++ < 5);

    IO.println(i);
 }

 -----------------------------------------------------------------------
```

```
 void main()
 {
   int i = 1;

   do
      {
     IO.println(i);
    }
   while ((i = i + 1) < 5);
 }
```

```
 ---------------------------------------------------------------------

 void main()
 {
   int i = 1;

   do
   {
    IO.print(i++ + " ");
   }
   while (i++ < 5);

   IO.println("\ni = " + i);
 }
```

---

2) while loop:

It is an **entry-controlled loop** because first, it will verify the **condition**, If Condition is **true** then only body will be executed.
It is preferable when number of iterations are **not fixed**.

Syntax:

```
while(condition)
{
  // statements
}
```

```
 While1.java
 ------------
 //Print  0 to -10
 void main()
 {
      int x = 0;

      while(x >= -10)
      {
           IO.print(x+"\t");
           x--;
      }

 }
```

```
 While2.java
 ------------
 void main()
 {
      while(false) //Unreachable
      {
           IO.println("Hello learner!!");
      }

      IO.println("Have you enjoyed it!!");

 }
```

```
 While3.java
 -----------
 void main()
 {
      boolean isValid = false;

      while(isValid)
      {
           IO.println("Hello learner!!");
      }

      IO.println("Have you enjoyed it!!");

 }
```

```
 While4.java
 -----------
 void main()
 {
      while(true)
      {
           IO.println("Hello learner!!");
      }

      IO.println("Have you enjoyed it!!"); //Unreachable

 }
```

```
 While5.java
 -----------
 void main()
 {
      final int x = 10;
       final int y = 20;

      while(x>y)
      {
           IO.println("x is greater than y");
      }

      IO.println("Out of the while loop");

 }
```